Network Threats and Mitigation & Physical Security and Risk

1) Create a table listing 5 common network threats (such as phishing, DDoS, man-in-the-middle, malware, and password attacks) and, for each, write one real-world example from an app or service you use (like WhatsApp, Instagram, or Paytm).

Threat and Example Table

- Phishing — Example: A fake Instagram or Paytm login message/link designed to trick a user into entering credentials.
- DDoS — Example: An online service such as a shopping or booking website can become unavailable when overwhelmed by malicious traffic.
- Man-in-the-Middle — Example: An attacker on an unsafe network attempts to intercept communication between a user and an online service.
- Malware — Example: A malicious file or application can infect a device and attempt to steal data or disrupt normal activity.
- Password Attacks — Example: Repeated guessing or credential-stuffing attempts against an online account such as Instagram or WhatsApp.

Mitigation

- Phishing — Verify links and messages, use MFA, and avoid entering credentials into suspicious pages.
- DDoS — Use traffic filtering, rate limiting, and dedicated DDoS protection.
- Man-in-the-Middle — Use HTTPS/TLS, secure Wi-Fi, and avoid untrusted networks.
- Malware — Keep software updated and use trusted security software.
- Password Attacks — Use strong unique passwords, MFA, and account lockout/rate-limiting controls.

2) Simulate a brute-force attack scenario by writing a simple Python script that tries to guess a 4-digit PIN code by checking all possible combinations against a preset value. The script should print the number of attempts it took to guess the correct PIN.

Step 1: Set the Preset PIN

```
secret_pin = "4729"
```

Step 2: Try Every 4-Digit Combination

```
secret_pin = "4729"
attempts = 0

for number in range(10000):
    guess = f"{number:04d}"
    attempts += 1

    if guess == secret_pin:
        print("PIN guessed:", guess)
        print("Number of attempts:", attempts)
        break
```

Explanation

- The loop checks values from 0000 through 9999 until the preset PIN is found.
- The attempts variable counts how many combinations were tested before the correct PIN was reached.
- This is a controlled classroom simulation against a locally defined test value, not a real account or system.

Expected Result

- For the preset PIN 4729, the script stops when 4729 is reached and prints the total number of attempts.

3) List 4 physical security risks that could affect a server room hosting a popular app like Zomato or BookMyShow, and suggest one mitigation step for each risk.

Physical Security Risks and Mitigation

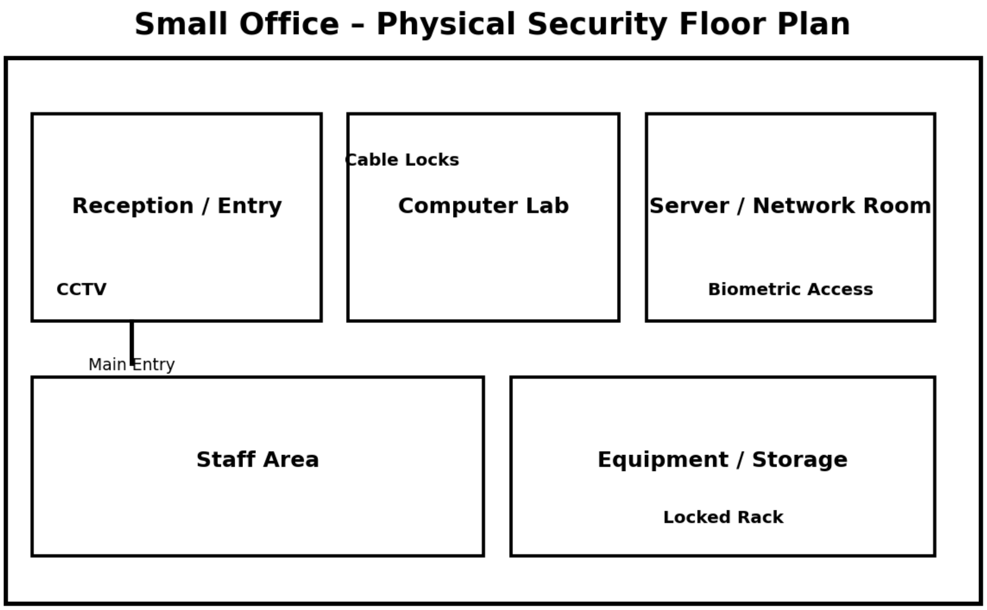
- Unauthorized physical access — Use controlled entry such as biometric access or an access card.
- Theft of servers/network equipment — Keep equipment in locked racks and restrict room access.
- Fire or overheating — Install smoke detection, fire suppression, temperature monitoring, and suitable cooling.
- Power failure — Use UPS systems, backup power, and power monitoring for critical equipment.

Additional Protection

- Maintain visitor logs and access records.
- Use CCTV monitoring around entrances and critical equipment areas.
- Keep network cables secured and protected from unauthorized tampering.

4) Draw a simple floor plan (hand-drawn or digital) of a small office and mark at least 3 physical security measures you would implement to protect network equipment (such as CCTV, biometric access, or cable locks).

Physical Security Floor Plan



- The reference floor plan below marks CCTV near the entry, biometric access for the server/network room, and cable locks/locked rack protection for network equipment.

Security Measures

- CCTV — Monitor the entrance and important equipment areas.
- Biometric Access — Restrict the server/network room to authorized personnel.
- Cable Locks / Locked Rack — Prevent unauthorized removal or tampering with network equipment.

5) Use ChatGPT or a similar AI tool to generate a checklist of 7 actions a network admin should take to mitigate risks from both network threats and physical security breaches, then review and edit the checklist to make it specific for a college computer lab environment.

College Computer Lab Security Checklist

- 1. Enable a firewall on lab computers and keep its rules appropriately configured.
- 2. Keep operating systems, browsers, antivirus/security software, and applications updated.
- 3. Use strong unique passwords and enable MFA for important administrator and cloud accounts where supported.
- 4. Separate student devices and sensitive administrative systems using suitable network segmentation or VLANs.
- 5. Restrict physical access to switches, routers, servers, and network cabinets.
- 6. Use CCTV, equipment locks, and visitor/access records to reduce physical security risks.
- 7. Maintain regular backups and an incident-response procedure for malware, account compromise, equipment theft, or network outages.

Review and Final Note

- The checklist combines network protection with physical security controls so that both cyber and physical risks are addressed.
- For a college computer lab, permissions should follow least privilege and sensitive network equipment should be accessible only to authorized staff.